Technical Summary:
Uncertainty Quantification for In-Context Learning of Large Language Models

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1 Research Problem and Motivation

In-context learning (ICL) provides a task description and a small set of task-relevant demonstrations in the prompt, enabling a large language model (LLM) to adapt at inference time without weight updates. Despite strong performance, LLM outputs exhibit trustworthiness failures such as hallucination and confidently incorrect predictions. Prior uncertainty quantification (UQ) methods for LLMs often return a single confidence score derived from variance or entropy across multiple responses, but they do not explain why uncertainty occurs under ICL, where predictions depend jointly on (i) the sampled demonstrations and (ii) the model-side decoding strategy and configuration. The central gap is the absence of a principled decomposition that attributes predictive uncertainty to these two sources, enabling diagnosis of whether a wrong and uncertain prediction is driven by demonstration quality (data-side variability) or by the model configuration and decoding procedure (model-side ambiguity). The success criterion is an uncertainty estimate that both (a) better ranks misclassified instances as high-uncertainty and (b) provides a meaningful decomposition into distinct uncertainty sources that align with targeted applications such as out-of-domain (OOD) demonstration detection.

2 Related Work

Uncertainty decomposition in machine learning commonly distinguishes Aleatoric Uncertainty (AU), arising from noise or variability in data, and Epistemic Uncertainty (EU), arising from model uncertainty such as inadequate capacity or parameter uncertainty. Classical decomposition techniques include Bayesian neural networks, deep ensembles, and Monte Carlo dropout. For language models, earlier work studied calibration and uncertainty measures for NLP tasks, and recent LLM-focused methods compute uncertainty from token-level likelihoods, token entropy, or semantic uncertainty. The compared baselines in this study include: (i) Likelihood-based Uncertainty, defined as normalized log-probability of generated tokens, (ii) Entropy-based Uncertainty, defined as entropy over generated token distributions, and (iii) Semantic Uncertainty, which clusters multiple generated sequences by semantic embeddings and computes entropy across clusters. These approaches treat the generated sequence broadly and do not explicitly separate the contribution of demonstration sampling from the contribution of model configuration and decoding. The proposed method differs by formulating ICL predictive uncertainty under a Bayesian perspective with a latent concept variable, and by decomposing uncertainty using mutual information, followed by an entropy approximation designed for free-form LLM outputs.

3 Dataset Construction

No new dataset is introduced. Experiments use established natural language understanding (NLU) benchmarks, and demonstrations are sampled from each dataset's training split (training-set sizes and preprocessing are not specified in the paper). Each benchmark is cast as a deterministic classification-style task, with prompts instructing the LLM to output only a numerical label.

Table 1: Datasets and task characteristics (all are classification-style NLU evaluations).

Dataset	Task	#Classes	#Test	Notes
EMOTION	Sentence-level emotion classification	6	2,000	Labels: sadness, joy, love, anger, fear, surprise.
Financial Phrasebank	Financial sentiment classification	3	850	Labels: negative, neutral, positive.
SST-2	Movie review sentiment classification	2	872	Labels: positive, negative.
CoLA	Linguistic acceptability classification	2	1,040	Predict grammatical vs ungrammatical.
AG_News	News topic classification	4	1,160	Labels: World, Sports, Business, Sci/Tech.

4 Query Protocol and Task Definitions

Prompt and ICL notation

The prompt is a concatenation of $T{-}1$ demonstrations and one test query: $x_{1:T} = [x_1, \ldots, x_{T-1}, x_T]$, where each demonstration contains a question and its labeled answer, and x_T is a test question without the answer. The model produces an output y_T for x_T. The paper introduces a latent concept variable z to represent the task concept inferred from the demonstrations, and a parameter/configuration variable Θ to represent model-side factors such as decoding strategy and decoding hyperparameters.

Demonstration sampling strategies

Two demonstration selection strategies are evaluated:

- **Random sampling:** demonstrations are sampled uniformly from the training set regardless of label.
- **Class-balanced sampling:** demonstrations are sampled to ensure at least one example per label class.

The number of demonstrations per dataset is selected to maximize task performance (exact selection rationale is not specified in the paper); the adopted counts are summarized in Table 2.

Evaluation tasks and success criteria

The primary evaluation treats uncertainty as a ranking signal for errors. Each test instance is assigned label 0 if classified correctly and 1 if misclassified. A successful uncertainty score ranks misclassified samples as higher uncertainty. Additional applications include:

Table 2: Number of demonstrations used per dataset and sampling rule.

Dataset	Random (#demos)	Class-balanced (#demos per class)
EMOTION	6	1 per class
Financial	6	2 per class
SST-2	4	2 per class
CoLA	2	1 per class
AG_News	4	1 per class

- **OOD demonstration detection:** distinguish in-domain demonstrations (label 0) from less relevant or OOD demonstrations (label 1) using uncertainty scores.
- **Semantic OOD (SOOD) test detection:** mask out several classes in the task description and treat masked-class test instances as OOD (label 1), expecting higher uncertainty on such instances.

Metrics

Task accuracy (ACC) is reported for classification. Uncertainty quality is evaluated by:

- **AUPR:** area under the Precision–Recall curve for separating misclassified (positive) from correct (negative) instances across thresholds.
- **AUROC (ROC):** area under the ROC curve; a random ranking yields 0.5, and an ideal ranking yields 1.

5 Modeling Approach

Bayesian formulation for ICL predictive distribution

The paper models ICL predictive uncertainty with a Bayesian perspective using a latent concept z and model configuration Θ:

$$p(y_T \mid x_{1:T}) \approx \int p(y_T \mid \Theta, x_{1:T}, z)\, p(z \mid x_{1:T})\, q(\Theta)\, dz\, d\Theta. \tag{1}$$

Purpose. Establish a probabilistic view in which uncertainty arises from both demonstration-induced latent concepts (z) and model configuration variability (Θ).
Symbols. $x_{1:T}$: prompt containing demonstrations and test query; y_T: model output for the test query; z: latent concept inferred from demonstrations; Θ: model parameters/configurations; $q(\Theta)$: an approximate posterior over Θ.
Intuition. Different demonstrations induce different inferred task concepts, and different decoding strategies and settings induce different outputs, so $p(y_T \mid x_{1:T})$ mixes these effects.
Usage. This formulation underpins the subsequent decomposition of total predictive uncertainty into EU (model-side) and AU (demonstration-side).

Uncertainty decomposition via mutual information

Total predictive uncertainty is measured by entropy. EU is the expected conditional entropy when conditioning on a specific configuration Θ and varying the latent concept. AU is captured by the mutual information between the output and latent concept under fixed Θ:

$$I(y_T, z \mid \Theta) = H(y_T \mid x_{1:T}, \Theta) - \mathbb{E}_z[H(y_T \mid x_{1:T}, z, \Theta)] \,. \tag{2}$$

Purpose. Define AU as the portion of uncertainty attributable to demonstration-induced latent concept variability, after accounting for EU.
Symbols. $I(\cdot, \cdot \mid \cdot)$: conditional mutual information; $H(\cdot)$: (differential) entropy; $\mathbb{E}_z[\cdot]$: expectation over latent concepts induced by demonstration sampling.
Intuition. If outputs change substantially across different latent concepts inferred from different demonstrations, then the output contains information about z, indicating large AU.
Usage. AU is computed as a difference between total entropy and expected conditional entropy, while EU corresponds to the conditional-entropy term.

Entropy approximation for free-form LLM outputs

Direct entropy over full generated sequences is impractical due to free-form generation and redundant tokens. The method extracts probabilities of answer-relevant token(s) and aggregates them into a label distribution. For each test query, repeated sampling over L demonstration sets yields a label-by-demonstration matrix $M \in \mathbb{R}^{K \times L}$, where K is the number of labels and column j stores the aggregated (unnormalized) answer probabilities for demonstration set j. Let $\sigma(\cdot)$ normalize a vector to a probability distribution. The paper approximates EU and AU as:

$$\mathrm{EU} = \frac{1}{L}\sum_{j=1}^{L} H(\sigma(M_{:,j}))\,, \qquad \mathrm{AU} = H\left(\sigma\left(\sum_{j=1}^{L} M_{:,j}\right)\right) - \frac{1}{L}\sum_{j=1}^{L} H(\sigma(M_{:,j}))\,. \tag{3}$$

Purpose. Provide a practical estimator of EU and AU for white-box LLMs that expose token probabilities.
Symbols. $M_{:,j}$: the jth column of M; $\sigma(\cdot)$: normalization operator; $H(\cdot)$: entropy; K: number of labels; L: number of demonstration sets.
Intuition. Each column represents uncertainty under a fixed demonstration set. Averaging column entropies yields EU. Aggregating across columns produces total uncertainty, and subtracting EU yields AU.
Usage. The estimator is computed per test instance. In practice, prompts instruct the model to output only the numerical label so that the answer token corresponds to the entire output, eliminating token-selection ambiguity.

Sampling model configurations through decoding

Sampling from $q(\Theta)$ by drawing model weights is not performed. Instead, beam search is used to generate multiple high-probability hypotheses, treated as an efficient alternative to sampling configurations, and the probabilities associated with beam hypotheses are used in the entropy approximation.

Minimal illustrative example

Consider a 4-class classification prompt with two demonstrations and one test query. Beam search with width $M = 3$ returns three decoded sequences. Suppose the extracted answer token probabilities assign mass to labels $\{0, 1, 2, 3\}$ as $(1.62, 0.81, 0.65, 0.00)$ after summing across the three beams. This becomes one column $M_{:,j}$ for demonstration set j. Repeating the procedure for $L = 3$ different demonstration sets produces $M \in \mathbb{R}^{4\times 3}$, after which Eq. (3) yields EU (average entropy across columns) and AU (additional entropy introduced by varying demonstrations).

Training, libraries, and environment

The method is inference-time only and does not fine-tune the LLMs. Optimization hyperparameters such as learning rate, batch size, optimizer, and training steps are Not specified in the paper. Software frameworks, library versions, hardware, and compute cost are Not specified in the paper.

6 Empirical Results

Experimental setup

Backbone models include LLaMA-2 chat models (7B, 13B, 70B) and OPT-13B for generalization analysis. Beam search uses beam size 10 and the maximum number of new tokens is 16. Demonstrations are sampled four times per dataset. Two demonstration strategies (Random, Class-balanced) are evaluated.

Main quantitative findings on misclassification ranking

Table 3 reports representative rows from the full comparison. The key pattern is that EU and AU outperform token-level baselines in most settings for identifying misclassified instances. For EMOTION with LLaMA-7B and Random demonstrations, EU achieves AUPR 0.688 and AUROC 0.667, exceeding Semantic Uncertainty (AUPR 0.598, AUROC 0.607) and token entropy (AUPR 0.448, AUROC 0.501). For Financial with LLaMA-70B and Class-balanced demonstrations, EU achieves AUPR 0.893 and AUROC 0.804, surpassing Semantic Uncertainty (AUPR 0.774, AUROC 0.649). For SST-2 with LLaMA-13B and Class-balanced demonstrations, EU attains AUROC 0.836 at AUPR 0.397, while AU attains AUROC 0.639 at AUPR 0.367, indicating that EU is the stronger indicator in that configuration.

Table 3: Representative uncertainty ranking results (higher is better). ACC is classification accuracy.

Dataset	Setting	ACC	Entropy AUPR	Entropy AUROC	Semantic AUPR	EU AUPR	EU AUROC
EMOTION	LLaMA-7B Random	0.407	0.448	0.501	0.598	0.688	0.667
EMOTION	LLaMA-7B Class	0.411	0.657	0.538	0.697	0.745	0.691
Financial	LLaMA-70B Class	0.537	0.623	0.439	0.774	0.893	0.804
SST-2	LLaMA-13B Class	0.928	0.113	0.439	0.343	0.397	0.836
CoLA	LLaMA-70B Class	0.852	0.397	0.588	0.397	0.389	0.727
AG_News	LLaMA-13B Class	0.685	0.359	0.413	0.411	0.429	0.654

Effects of demonstration sampling and model scale

Class-balanced demonstrations generally increase AUPR and AUROC relative to Random sampling, attributed to reduced sampling bias by ensuring label coverage. Larger models tend to increase both ACC and uncertainty-ranking performance, but monotonic improvements are not guaranteed. For EMOTION, LLaMA-70B sometimes underperforms LLaMA-13B on specific uncertainty metrics, indicating that uncertainty behavior depends on task context rather than only scale.

Generalization across backbone models

On EMOTION, OPT-13B and LLaMA-2-13B show similar ROC patterns. In the reported comparison, EU AUROC improves from 0.55 (OPT-13B) to 0.59 (LLaMA-2-13B), while AU AUROC is 0.68 for both, indicating that EU reflects backbone strength more directly in this analysis.

Out-of-domain demonstrations and sensitivity of AU

Misclassification AUROC under OOD demonstrations on EMOTION shows that AU is more sensitive to demonstration relevance. With LLaMA-13B and Class-balanced sampling, AU AUROC decreases from 0.599 (Original demonstrations) to 0.524 (Relevant but non-identical demonstrations) and to 0.497 (OOD demonstrations from CoLA), a 17.0% drop from Original to OOD. In contrast, EU AUROC changes are smaller (from 0.686 to 0.673 in the same comparison), supporting the paper's claim that AU captures demonstration-induced instability more strongly.

OOD demonstration detection

For distinguishing in-domain versus OOD demonstrations on EMOTION with LLaMA-2-13B, EU provides the strongest indicator. For the OOD-demo scenario, EU achieves AUPR 0.784 and AUROC 0.941, compared to Semantic Uncertainty (AUPR 0.698, AUROC 0.712) and AU (AUPR 0.773, AUROC 0.607). This demonstrates that decomposition improves not only misclassification ranking but also a targeted diagnostic task where the cause is specifically tied to demonstrations.

Semantic OOD (SOOD) test detection

When two classes are masked from the EMOTION label set and masked-class test instances are treated as SOOD, EU remains the best indicator. For LLaMA-2 7B, EU achieves AUPR 0.548 and AUROC 0.658, exceeding Semantic Uncertainty (AUPR 0.477, AUROC 0.532). For LLaMA-2 13B, EU achieves AUPR 0.525 and AUROC 0.592. AU does not improve in this setting, consistent with the paper's explanation that AU does not necessarily increase under inappropriate task descriptions and demonstrations when semantic shifts dominate.

Observed weaknesses relative to objectives

The decomposition yields two scores, but EU and AU do not dominate uniformly across all datasets and settings. Some metrics on specific datasets show non-monotonic trends with model size, and AU can be a weaker indicator than EU for misclassification ranking in several settings, indicating that demonstration-induced variability is not always the primary driver of errors. The baseline methods can also be competitive on certain tasks and metrics, particularly when token-level uncertainty aligns well with error patterns.

7 Summary

A Bayesian latent-variable view of ICL is used to formalize predictive uncertainty as arising jointly from demonstration-induced latent concepts and model-side configuration variability. Mutual-information-based decomposition separates AU and EU, and an entropy approximation is introduced to handle free-form LLM outputs by extracting answer-relevant token probabilities and aggregating them into label distributions. Experiments on five NLU benchmarks with LLaMA-2 (7B, 13B, 70B) and OPT-13B show that EU and AU typically improve AUPR and AUROC for ranking misclassified instances relative to likelihood-, entropy-, and semantic-based baselines. The decomposition further enables diagnostic applications, including OOD demonstration detection where EU achieves AUROC 0.941 and AUPR 0.784 in a reported setting, and misclassification robustness studies showing AU sensitivity to demonstration relevance with up to a 17.0% AUROC decrease under OOD demonstrations. The paper states two limitations: applicability is focused on NLU tasks such as multiple-choice question answering, text classification, and linguistic acceptability, and the entropy-based estimator has limited usage for generation tasks because semantic importance within generated sequences is not identifiable under the proposed procedure.